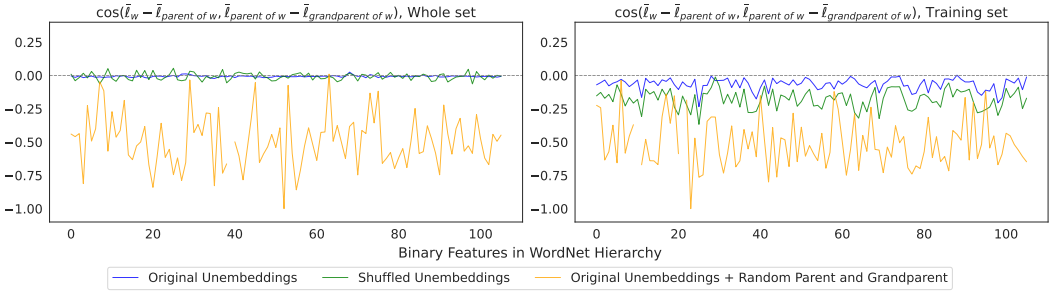

Figure 16: WordNet **verb** hierarchy is encoded as the orthogonal structure predicted by statement (d) in Theorem 8.

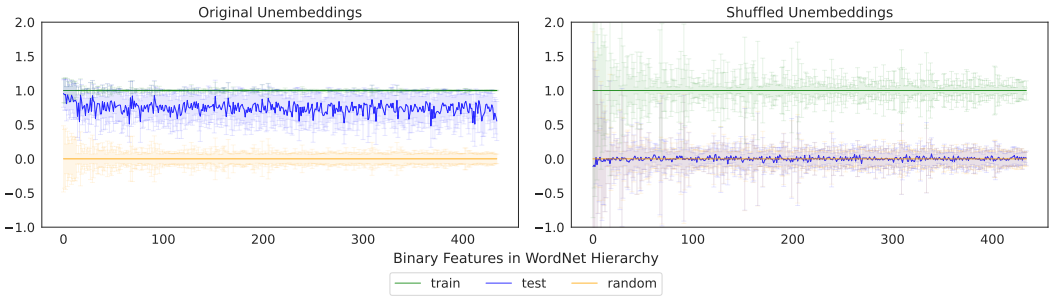

Figure 17: Vector representations for most binary features in the WordNet noun hierarchy exist in the **LLaMA-3** model.

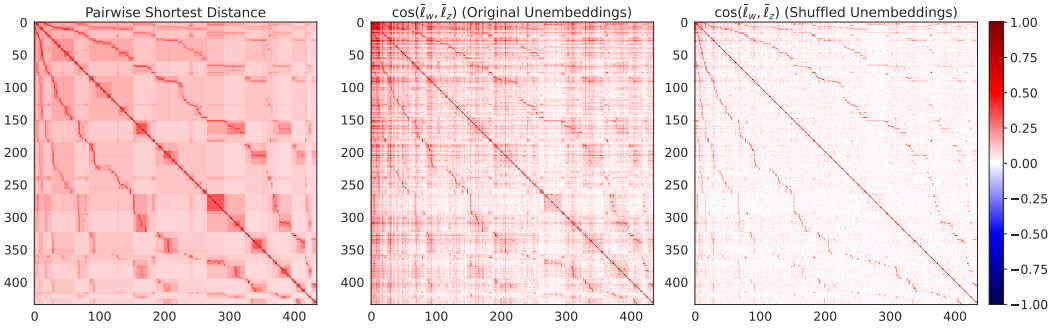

Figure 18: WordNet noun hierarchy is encoded in **LLaMA-3** representation space.

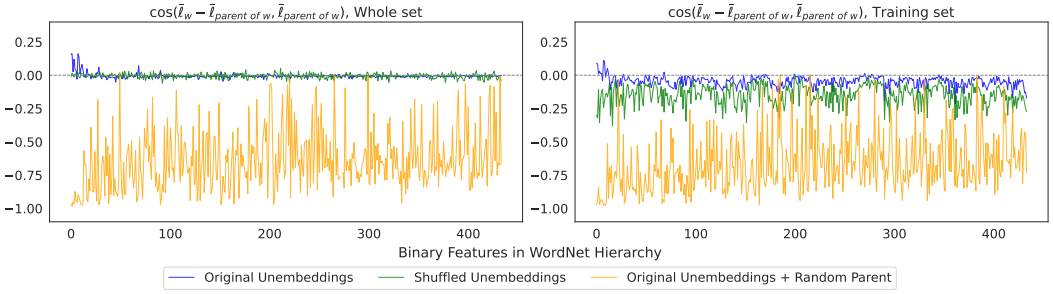

Figure 19: In **LLaMA-3** representation space, WordNet noun hierarchy is encoded as the orthogonal structure predicted by statement (a) in Theorem 8.

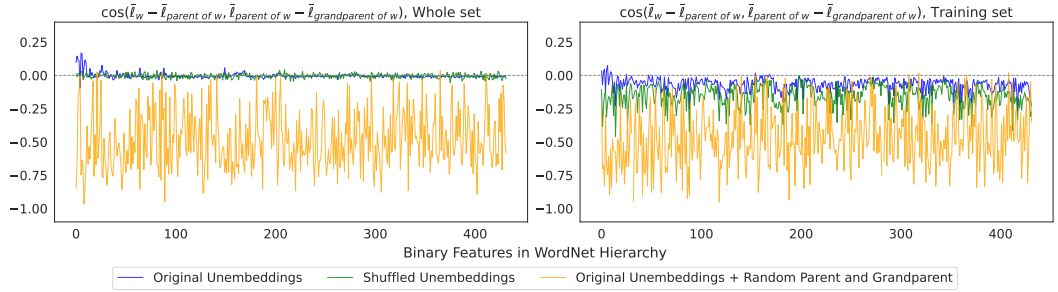

Figure 20: In **LLaMA-3** representation space, WordNet noun hierarchy is encoded as the orthogonal structure predicted by statement (d) in Theorem 8.